An analytic toy for the shape of $D_\ell^{\rm cross}(x_{\rm HII})$

kSZ$^2\times$galaxy cross-power during reionisation: what sets the curve?

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1 Question

La Plante, Sipple & Lidz (2022) Fig. 7 plots $D_\ell^{\rm cross}$ (cross-power of the filtered kSZ^2 map with a galaxy window of width $\Delta z = 1$) against the mean ionised fraction $x_{\rm HII}$ of the window. It rises from $x_{\rm HII} \simeq 0$, peaks near $x_{\rm HII} \simeq 0.1$–0.2 and has fallen to almost nothing by $x_{\rm HII} \simeq 0.8$. Is that shape already explained by a few lines of algebra, and which ingredient makes it?

2 The model

Estimator. $S = [F\Theta]^2$ with $\Theta = \sum_i \theta_i$, $\theta_i = -W_i(1+\delta_i)x_{e,i}v_i/c$, $W_i = \tau'(1+z)^2 e^{-\tau_i} ds_i$. The toy treats the lightcone as independent slices (the "diagonal", square-then-project ordering) and works in the squeezed limit: the small-scale kSZ^2 power $s(z)$ of a slice is modulated by the long-wavelength density δ_L of that slice, $s \to s\,(1 + R\,\delta_L)$, and the galaxy map is $g = \int w\, b_g\, \delta_m$.

Slice power. With $K(z) = \tau_{\rm pref}(1+z)^2 e^{-\tau(z)}$ and $v(z) = E(z)\,\Omega_m(z)^{0.55} D(z)/(1+z)$ (i.e. $aHfD$ in units of H_0),

$$s(z) = K^2 v^2 \left[\bar{x}^2 + \epsilon\,\bar{x}(1-\bar{x})\right]. \tag{1}$$

The first term is the kSZ of a uniformly ionised slice with fraction $\bar{x}(z)$ moving with the velocity field. The second is the patchy term: a binary ionisation field has variance $\bar{x}(1-\bar{x})$, and ϵ is the ratio of patchy to homogeneous power at the filter scale. *ϵ is the only shape parameter.*

Response to a long mode ($\partial s/\partial\delta_L = R\,s$):

$$\text{density/velocity:}\quad (1+\delta)^2 \to 2\delta_L \;\Rightarrow\; R_{\rm d}\,s = (2+r_v)\,K^2 v^2\left[\bar{x}^2 + \epsilon\bar{x}(1-\bar{x})\right], \quad r_v = 0, \tag{2}$$

$$\text{ionisation (zreion):}\quad z_{\rm re} = z_{\rm mean} + (1+z_{\rm mean})\,b_{zm}(k)\,\delta_L(z_{\rm mean}), \tag{3}$$

$$\frac{\partial\bar{x}}{\partial\delta_L} = (1+z_{\rm mean})\,b_{zm}(k)\,\frac{D(z_{\rm mean})}{D(z)}\,p(z), \qquad p = -\frac{d\bar{x}}{dz}, \tag{4}$$

$$R_{\rm i}\,s = K^2 v^2\left[2\bar{x} + \epsilon(1-2\bar{x})\right]\frac{\partial\bar{x}}{\partial\delta_L}. \tag{5}$$

Here $b_{zm}(k) = b_0/(1+k/k_0)^{0.2}$ with $b_0 = 1/1.686$, $k_0 = 0.9\,h\,{\rm Mpc}^{-1}$, $z_{\rm mean} = 8$ (La Plante Eqs. 1–3, as in `zreion/box.py`).

Cross-power (Limber, top-hat galaxy window of width Δz, $k = \ell/\chi$):

$$D_\ell = \frac{\ell(\ell+1)}{2\pi}\,\frac{A}{\Delta z}\int_{z_0-\Delta z/2}^{z_0+\Delta z/2} dz\;(R_{\rm d}s + R_{\rm i}s)\;b_g(z)\,\frac{D(z)^2 P_{\rm lin}(\ell/\chi)}{\chi^2}, \tag{6}$$

with $P_{\rm lin}$ Eisenstein–Hu, $b_g = 2.1(1+z) - 5.3$ (the BlueTides fit used by the repo; $\simeq 9.4$ at $z = 6$, 22 at $z = 12$) for every tracer, and one overall amplitude A per ℓ (it absorbs $T_{\rm CMB}^2$, the filter's share of the slice power and the velocity normalisation). The $x_{\rm HII}$ of a window is $\langle\bar{x}\rangle$ over it. Simplification: the power is taken per unit redshift; the factor $d\chi/dz$ is omitted (including it leaves the best ϵ unchanged, misfit $0.33 \to 0.43$).

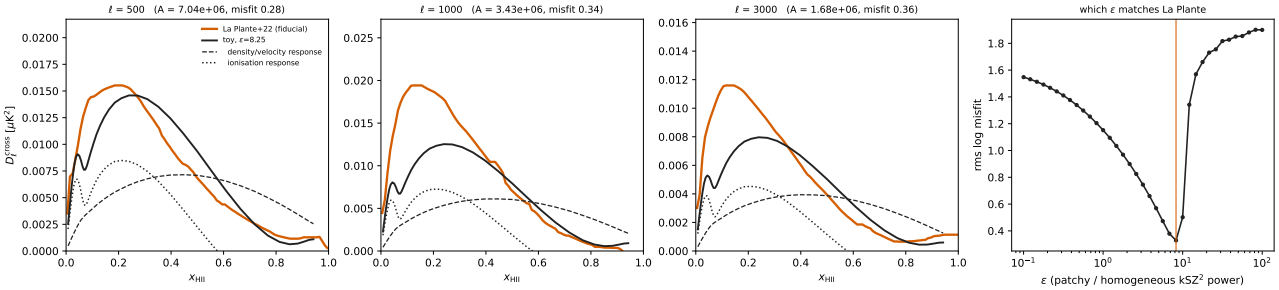
History. La Plante's digitised fiducial history, made monotonic, Gaussian-smoothed ($\sigma_z = 0.3$) so that $p = -d\bar{x}/dz$ is usable, with $\bar{x} = 1$ below the table and an $e^{-(z-11.86)}$ tail above its end ($\bar{x} = 0.06$). $\tau(z) = \int \tau_{\rm pref}\,\bar{x}\,(1+z)^2 d\chi$ from $z = 0$.

z	6	8	9	10	11	12
$\bar{x}$	0.91	0.44	0.24	0.13	0.08	0.05
τ	0.046	0.062	0.066	0.068	0.070	0.071
K^2v^2 $[10^{-10}]$	0.42	0.87	1.2	1.6	2.0	2.6
$p = -d\bar{x}/dz$	0.10	0.26	0.15	0.07	0.03	0.04
b_g	9.4	13.6	15.7	17.8	19.9	22.0
$D(z)$	0.18	0.14	0.13	0.12	0.11	0.10

Redshift functions entering the integrand (fiducial history).

3 Fit

For each ϵ and each $\ell \in \{500, 1000, 3000\}$ the amplitude A is the log-space least-squares value against La Plante's digitised Fig. 7 curve over $x_{\rm HII} = 0.1$–0.9 (40 points); the misfit is the rms of $\ln D_{\rm LP} - \ln(A\, D_{\rm toy})$. ϵ is the value minimising the misfit averaged over the three ℓ.



Orange: La Plante (fiducial). Black: toy ($\epsilon = 8.25$), with its density/velocity part (dashed) and ionisation part (dotted). Right: misfit versus ϵ.

4 Results

- **Best** $\epsilon \simeq 8$ (misfit 0.28/0.34/0.36 in log at $\ell = 500/1000/3000$; $A = 7.0, 3.4, 1.7\times10^6$). Peak at $x_{\rm HII} \simeq 0.24$ (La Plante: 0.12–0.19); falls to ~ 0 by $x_{\rm HII} \simeq 0.8$. The shape is therefore reproduced to within the digitisation and the toy's crudeness; the peak sits slightly too late.
- For $\epsilon \lesssim 3$ (homogeneous velocity term competitive) the misfit is $\gtrsim 0.7$: the peak moves to $x_{\rm HII} \simeq 0.3$ ($\epsilon = 3$), 0.55 ($\epsilon = 1$), 0.7 ($\epsilon = 0$) and the curve keeps 30–70% of its peak value at $x_{\rm HII} = 0.9$, whereas La Plante keeps $\sim 2\%$ (toy at $\epsilon = 8$: 6%).
- The *density/velocity* part alone peaks at $x_{\rm HII} \simeq 0.4$–0.45 and cannot fall fast enough. The *ionisation* part alone peaks at $x_{\rm HII} \simeq 0.2$ and goes to zero where $2\bar{x} + \epsilon(1 - 2\bar{x})$ changes sign, $\bar{x} = \epsilon/[2(\epsilon - 1)] \simeq 0.57$; this, together with $p \to 0$ as reionisation ends, produces the late fall.
- Robustness: $z_{\rm mean} = 7, 8, 9$ give the same ϵ and misfit, because $(1 + z_{\rm mean})D(z_{\rm mean})$ is nearly constant. The minimum in ϵ is sharp (misfit 0.33 at 8.25, 0.50 at 10, 1.0 at 11, ≥ 1.3 for $\epsilon \geq 12$) because the total goes through zero; ϵ should not be quoted as a measurement.

5 What it says

The early rise and the late fall of $D_\ell^{\rm cross}(x_{\rm HII})$ are produced by *patchy* ionisation: the kSZ^2 power of a slice that is partly ionised (growing as $\bar{x}(1 - \bar{x})$) is correlated with the large-scale density mainly through where bubbles form (the zreion response $\propto p(z)$), not through the homogeneous velocity term. This is consistent with the component study, where $v_{\rm los}$ and the ionisation morphology both follow the kSZ curve. A tracer that is a better proxy for δ_L at the patchy epoch (larger effective b_g, less noise) therefore carries more of this signal.

6 Caveats and checks to do

- Squeezed-limit, slice-diagonal toy: no cross-slice pairs, no LOS cancellation of the homogeneous term. $\epsilon \sim 8$ may partly stand in for that cancellation, so it is a statement about the relative size of the patchy

and homogeneous terms in the *estimator*, not a measured physical ratio.

- $x_{\rm HII} < 0.1$ is controlled by the first points of the digitised history (a small wiggle remains); $b_g(z)$ is one shape for all tracers; only the shape is fitted, the amplitude is not predicted.
- Not yet done: the same overlay on our own measured curves (`--ours-root`) and a direct test of ϵ from the simulation (power of the x_e-only versus v-only kSZ^2 maps per window, scripts 48/49).

7 Reproduce

```
python scripts/50_toy_model.py --out-dir data/plots/2026-10-23_toy \
   [--ours-root data/plots/2026-10-16_tracer_cross] [--eps 8.25]
```